

Containerisation: Containerising the applications helps in making them move across public, private, and on-premises clouds, and must be looked at before moving to a hybrid cloud.

Cloud interoperability: Integration between several cloud offerings across multiple cloud service providers and cloud types is key to the success of hybrid cloud adoption.

Figure 2 lists six drivers of hybrid cloud adoption. These are:

Pace of cloud adoption: A big bang approach is not going to fly. Factors that will help decide the rate of cloud adoption are the complexity of the application, data requirements, regulatory and compliance needs, modernisation prerequisites, cost implications, etc.

Complex integrations: There is always a need to connect an application to historical data that resides on the on-premises servers, even though the migration to cloud has happened. Businesses must invest in a good integration strategy before moving to a hybrid cloud. Applications with high integration may need to be co-located to deliver the best experience. Point of delivery applications may be deployed on specialised hardware systems or integrated with machinery that requires applications to be hosted on-premises.

Security and regulatory requirements: Applications with regulatory and compliance requirements need some of the data to reside on-premises. Due diligence is needed to select the right applications for the hybrid cloud.

Best-of-breed environments: The best environment must be chosen for the applications that require special hardware to deliver the functionality at the most optimal cost and effort.

High availability: Applications should be redundantly deployed for high availability across many cloud types to provide business process continuity and disaster recovery.

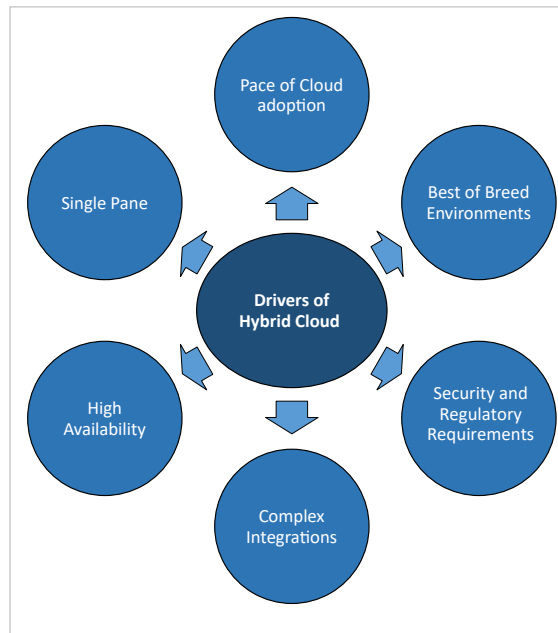


Figure 2: Drivers of hybrid cloud

Single pane, single view: Single pane, single view of all the applications across different cloud models is a must to ensure effective data management, security, and reduced costs.

Hybrid cloud management model

The key components of the hybrid cloud management model (Figure 3) are:

Channels: These are the various mediums through which users interact with the cloud, such as the web and IoT devices.

Enterprise application portfolio: As a first step, the applications that run on the hybrid cloud are categorised. Some of the core business applications are ERP, CRM, and HR systems. Factors like storage, latency, application dependencies and operational complexity need to be considered to decide on cloud adoption. Applications that change/evolve frequently are best moved to the cloud to leverage automated deployment. The next step is to decide on the need to redesign or re-architect an application to use it in the cloud.

Integrated enterprise cloud service management layer: This layer

helps in handling all the service requirements, delivering and managing cloud services across the enterprise. It includes all the cloud platforms deployed by the enterprise including public cloud, private cloud, third party hosted environments and on-premises legacy data centres. The policies and frameworks address interconnectivity and integration across multiple cloud components.

- **API gateway:** This manages and routes API requests, ensuring security and scalability.

- **Service gateway:** This focuses on managing microservices.
- **Adaptors/connectors:** These enable integration with various external systems and services.

Enterprise cloud management platform:

This platform provisions infrastructure components, platform services components, security and access management, and multi-cloud governance. It offers a self-service catalogue for users that helps them manage their services.

Enterprise container management layer: Containerising the solution makes the services cloud- and platform-agnostic. Container platforms are now available as a service (CaaS) which is robust and highly available, making multi-region deployment possible easily.

Public cloud: Public clouds offer the flexibility of deployment across multiple regions and for varying user loads. They typically have DMZ (demilitarized) and non-DMZ environments with perimeter security implemented through cloud native services. AWS, Azure, and GCP are major public cloud providers.